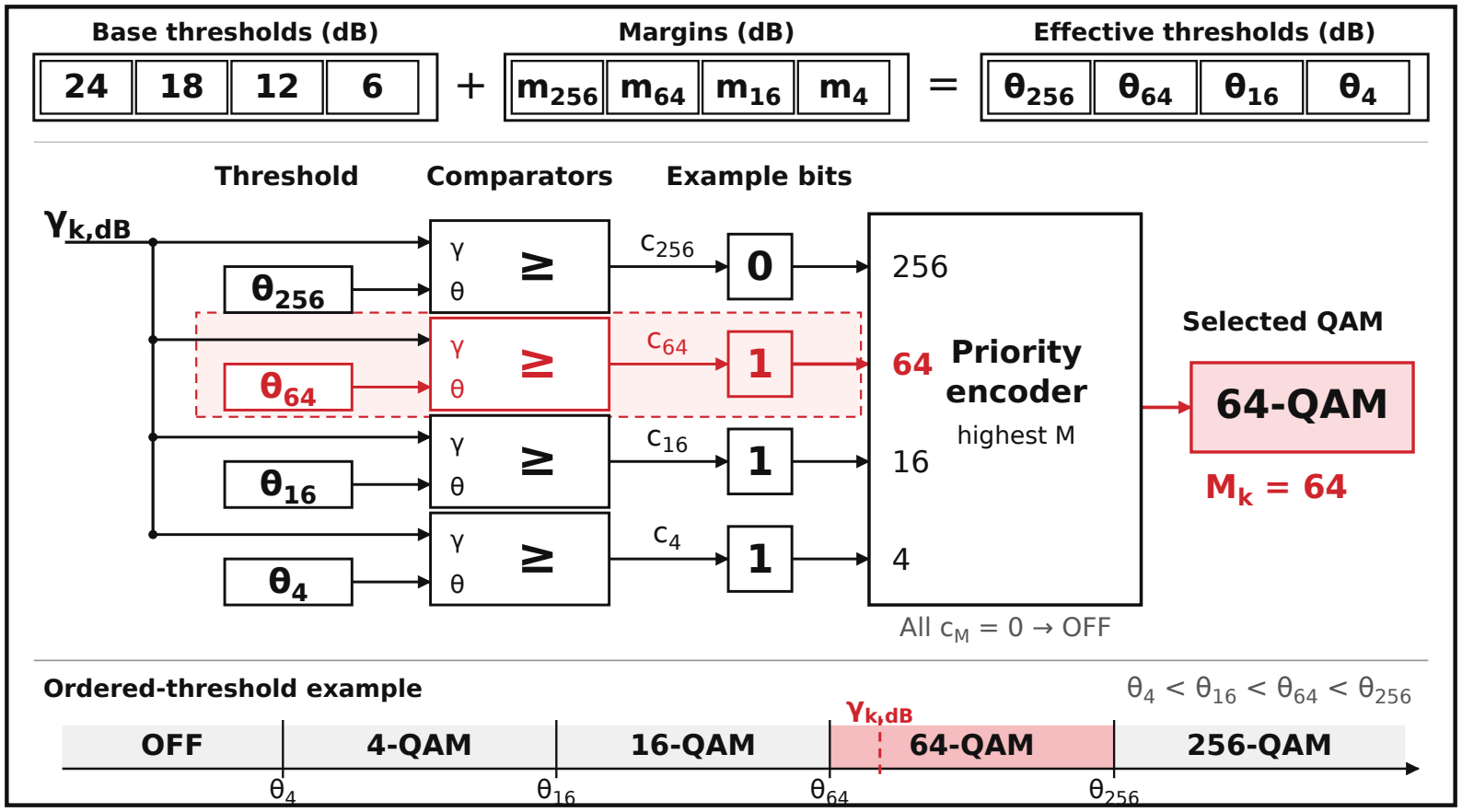

Base thresholds (dB)
24
18
12
6
+
Margins (dB)
$m_{256}$
$m_{64}$
$m_{16}$
$m_4$
=
Effective thresholds (dB)
$\theta_{256}$
$\theta_{64}$
$\theta_{16}$
$\theta_4$
Threshold
Comparators
Example bits
$\gamma_{k,dB}$
$\theta_{256}$
$\theta_{64}$
$\theta_{16}$
$\theta_4$
γ
θ
≥
$c_{256}$
$c_{64}$
$c_{16}$
$c_4$
0
1
1
1
256
64
16
4
Priority encoder
highest M
All $c_M = 0 \rightarrow$ OFF
Selected QAM
64-QAM
$M_k = 64$
Ordered-threshold example
$\theta_4 < \theta_{16} < \theta_{64} < \theta_{256}$
$\gamma_{k,dB}$
OFF
4-QAM
16-QAM
64-QAM
256-QAM
$\theta_4$
$\theta_{16}$
$\theta_{64}$
$\theta_{256}$